

SIMFLOW

Step-by-Step Process Documentation

Prototype Workflow + Azure Batch Execution Workflow

Team	T177
Members	Ch. Sri Neha • A. Keerthi • S. Bhavya Sri • K. Sri Sai Pavan
Project	SimFlow + Azure Batch Engineering Simulation Workload

1. SimFlow Prototype Process

This section records the process visible in the local SimFlow prototype, from opening the dashboard through execution monitoring and completion.

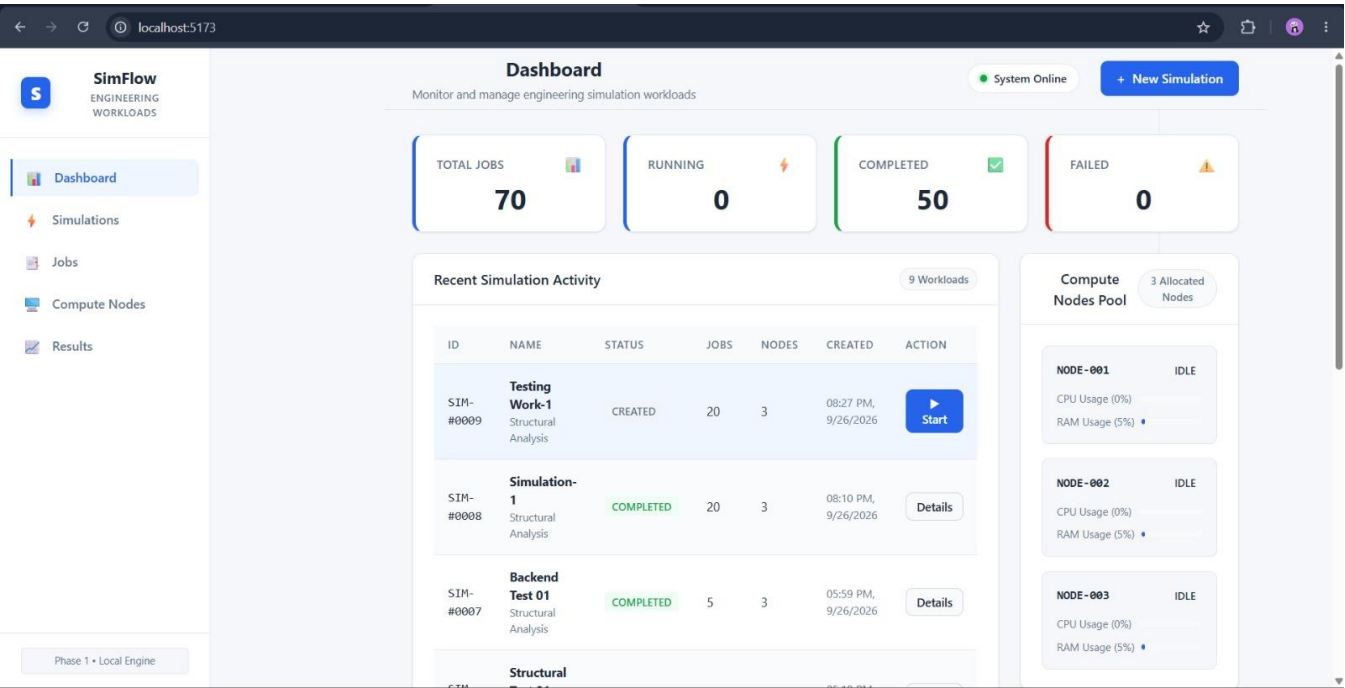


Figure 1. SimFlow dashboard — starting point for workload monitoring.

1

Open the SimFlow Dashboard

Launch the local SimFlow application. The Dashboard acts as the main control screen and provides access to Simulations, Jobs, Compute Nodes, and Results. It also summarizes total jobs, running jobs, completed jobs, and failed jobs.

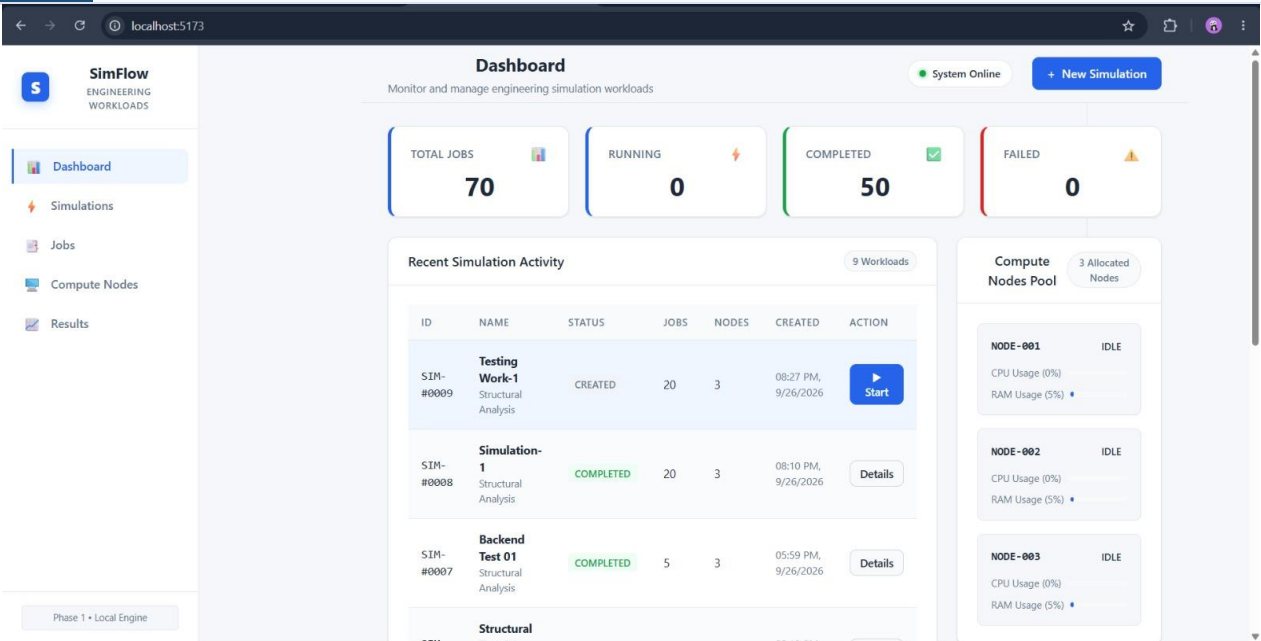


Figure 2. Dashboard view showing workload counts and the compute-node pool.

2

Create / Select a Simulation Workload

Use the Simulations area and the New Simulation action to define a workload. The prototype represents the workload as a simulation record that can later be executed as a set of jobs. The configuration includes the simulation type, compute-node requirement, priority, and workload details.

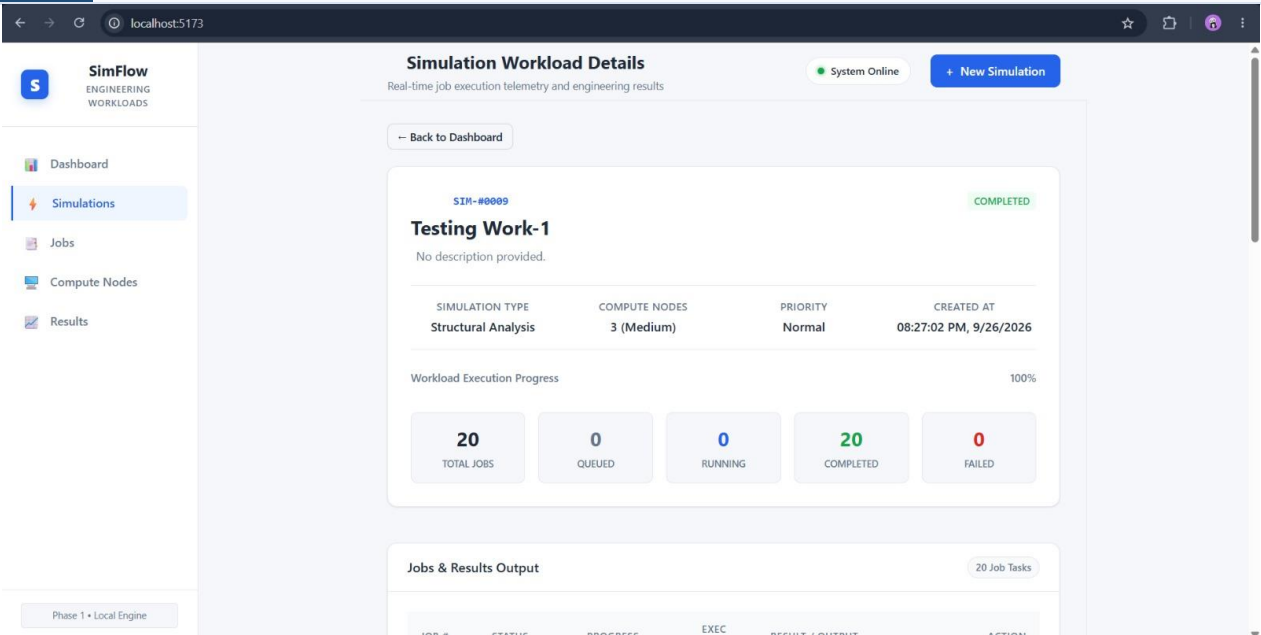


Figure 3. Simulation Workload Details page showing simulation type, nodes, priority and job counts.

3

Start the Simulation

Start the created simulation from the dashboard. After execution begins, the prototype changes the selected workload from a created/ready state to RUNNING and the dashboard reflects that a workload is actively executing.

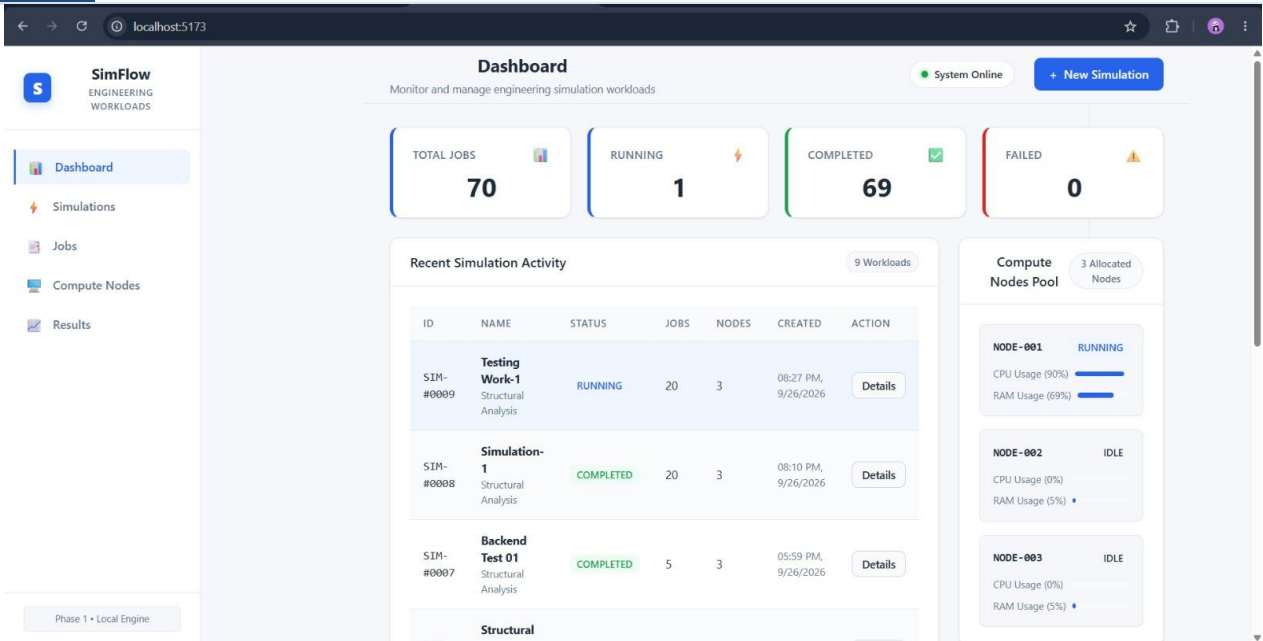


Figure 4. Running state: one workload is active and the first compute node shows CPU/RAM activity.

4

Monitor Execution

During execution, monitor the workload from the dashboard. The screen provides the current number of running and completed jobs, the state of individual workloads, and basic CPU/RAM indicators for the allocated nodes. This allows the user to follow the workload without opening each task separately.

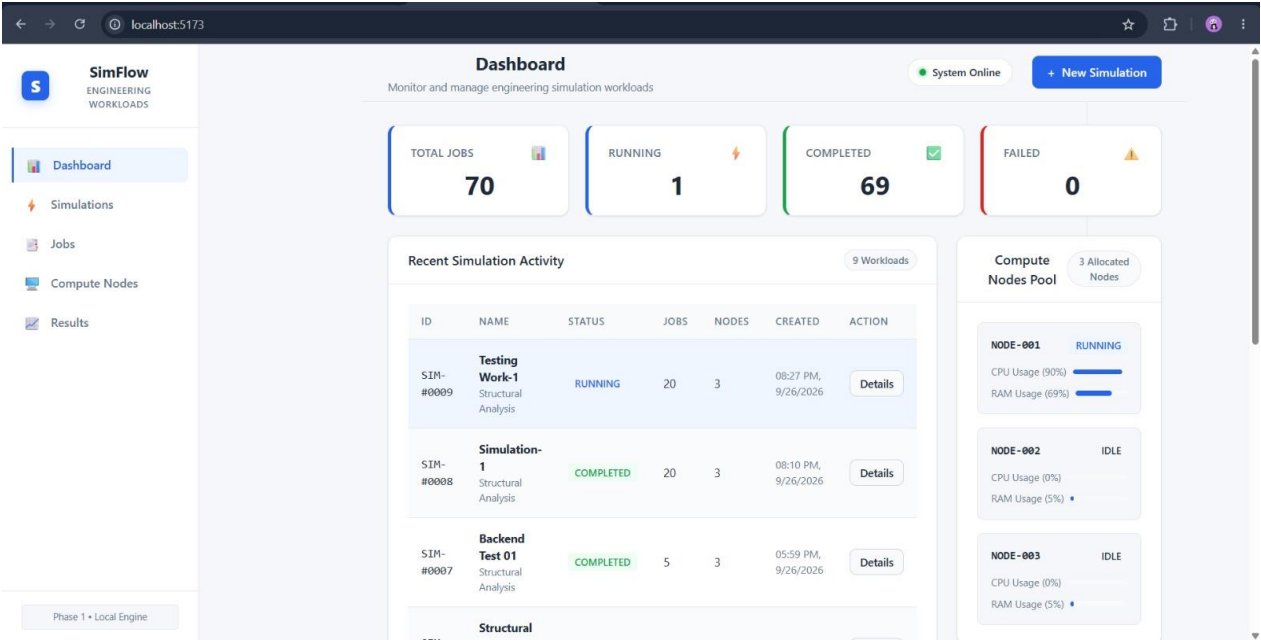


Figure 5. Prototype execution monitoring with job status and node utilization.

5

Complete the Workload

When the workload finishes, the running count returns to zero and the completed count increases. The simulation entry changes to COMPLETED, while the compute nodes return to an idle state.

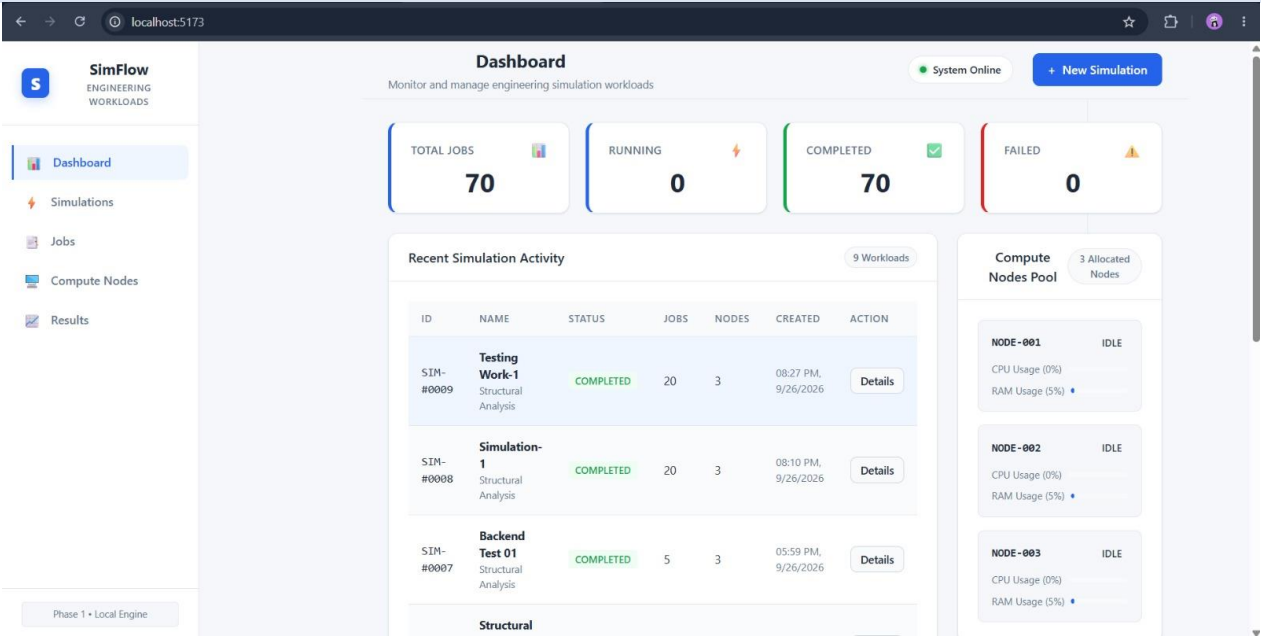


Figure 6. Completed workload state visible on the SimFlow dashboard.

6

Open Simulation Details and Review Results

Open the simulation details page to confirm the final execution state. The page shows the workload execution progress, total jobs, queued/running/completed/failed counts, and a Jobs & Results Output area for reviewing individual job results.

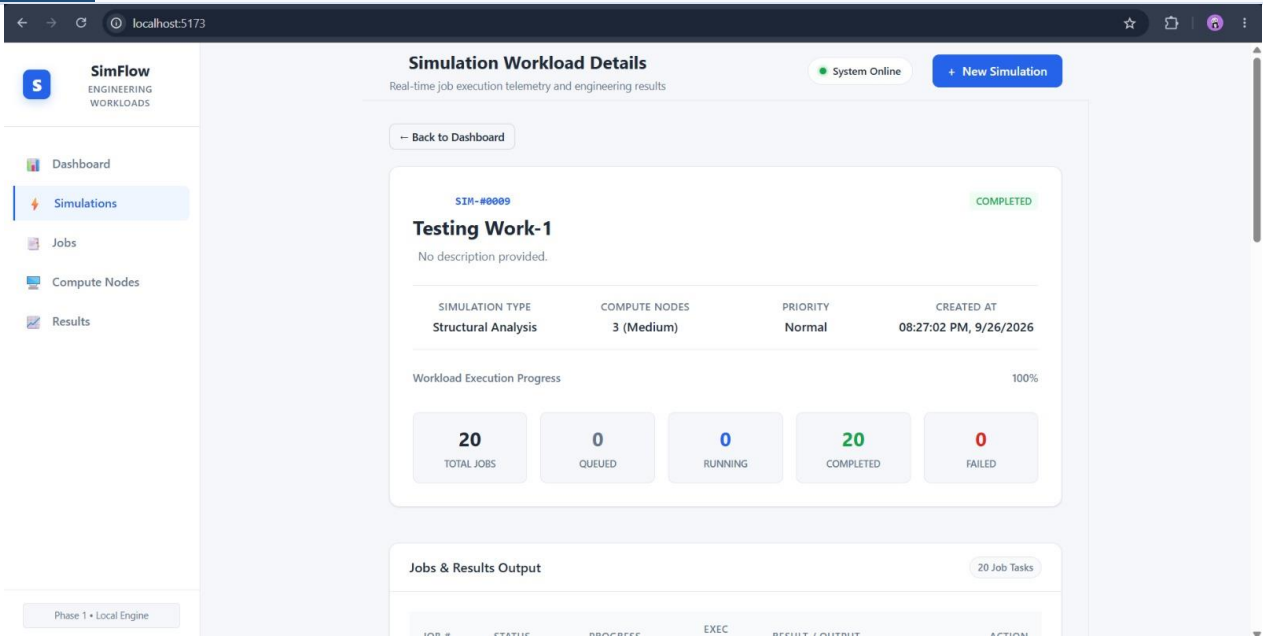


Figure 7. Completed simulation details with execution progress and job/result area.

2. Azure Batch Process

This section records the cloud-side process used to configure Azure Batch and execute the numerical simulation workload as independent tasks.

1

Open Azure Portal and Locate Project Resources

Sign in to the Azure Portal and locate the project resources. The project uses a Resource Group to organize the cloud components, together with the Azure Batch account and supporting resources used for the workload.

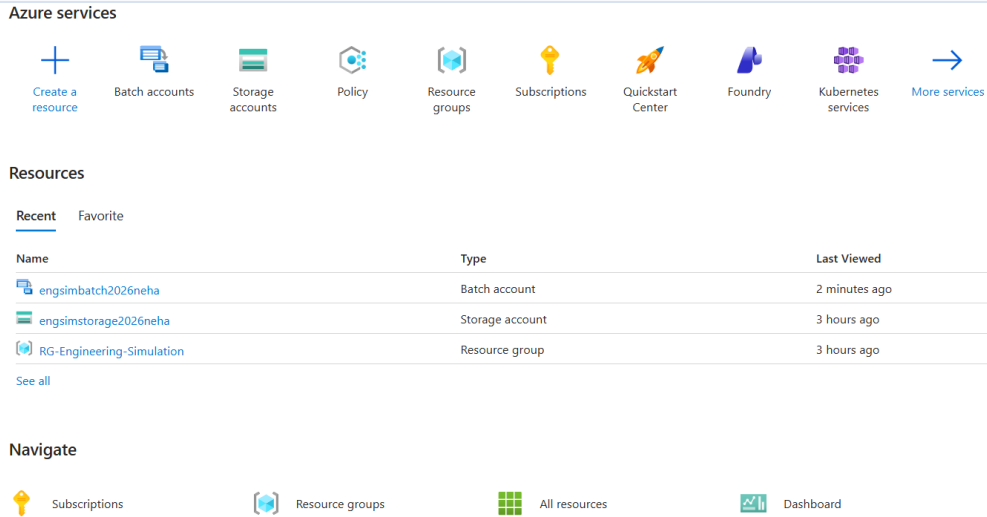


Figure 8. Azure portal resources used for the project.

2

Open the Azure Batch Account

Open the Batch account used for the workload. The account is the control point for managing compute pools, jobs, tasks, scheduling, and execution status. In this project the account shown in the supplied evidence is engsimbatch2026neha.

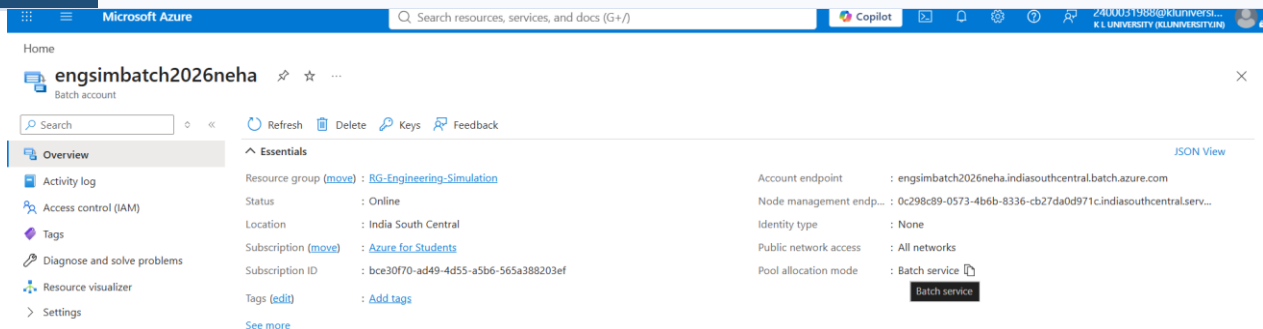


Figure 9. Azure Batch account overview.

3

Create / Verify the Compute Pool

Open the Pools section and verify the pool configuration. The supplied implementation uses the pool simulation-pool with two dedicated Standard_D2s_v3 nodes, providing the compute capacity used for the demonstration.

The screenshot shows the Azure Batch 'Pools' page for the account 'engsimbatch2026neha'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Features, Applications, Pools (selected), Jobs, Job schedules, Monitoring, Automation, and Help. The main content area has a search bar, '+ Add', '+ Add (JSON editor)', 'Columns', and 'Refresh' buttons. Below these are filters for 'State == all' and 'Add filter'. A 'Search for pool by ID or display ...' input field is present. A 'Pagination effort limit' is set to 1. The table below lists the pool configuration:

Pool ID	Dedicated nodes	Spot/low-priority nodes	Current vCPUs	VM size	Allocation state
simulation-pool	2	0	4	STANDARD_D2S_V3	Steady

Figure 10. simulation-pool configuration in Azure Batch.

4

Create the Batch Job

Create the logical job that groups the simulation tasks. The supplied Azure implementation uses the job ID simulation-job and links it to simulation-pool.

The screenshot shows the Azure Batch 'Jobs' page for the account 'engsimbatch2026neha'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Features, Applications, Pools, Jobs (selected), Job schedules, Monitoring, Automation, and Help. The main content area has a search bar, '+ Add', '+ Add (JSON editor)', 'Columns', and 'Refresh' buttons. Below these are filters for 'State == all' and 'Add filter'. A 'Filter by ID or pool' input field is present. A 'Pagination effort limit' is set to 1. The table below lists the job configuration:

ID	State	Pool	Created
simulation-job	Active	simulation-pool	26 Sept 2026, 22:56:55

Figure 11. simulation-job linked to simulation-pool.

5

Submit the Simulation Tasks

Add the individual simulation tasks to the Batch job. The project workload contains simulation-task-001 through simulation-task-300. Each task represents one independent numerical computation that can be scheduled to an available compute node.

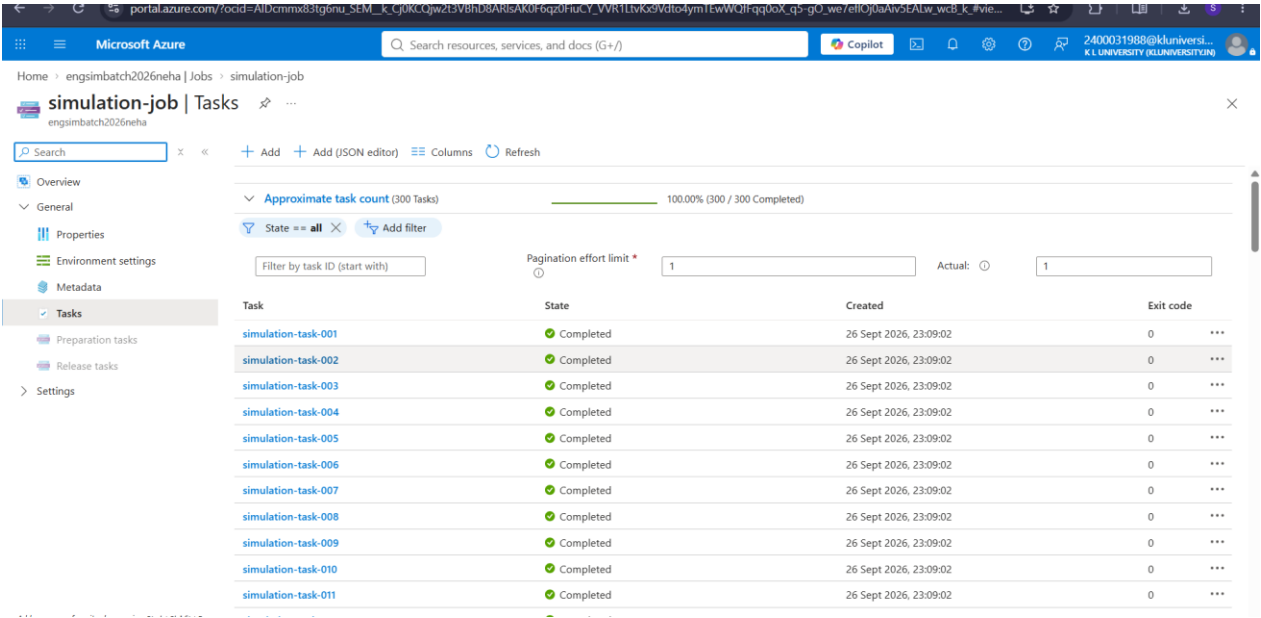


Figure 12. Azure task list showing the submitted workload.

6

Schedule and Execute Tasks on the Pool

Azure Batch places queued tasks onto available compute nodes in the pool. Because the tasks are independent, multiple tasks can execute concurrently when sufficient compute capacity is available. The two-node pool keeps the educational demonstration small and resource-conscious.

Task	State	Created	Exit code
simulation-task-001	Completed	26 Sept 2026, 23:09:02	0
simulation-task-002	Completed	26 Sept 2026, 23:09:02	0
simulation-task-003	Completed	26 Sept 2026, 23:09:02	0
simulation-task-004	Completed	26 Sept 2026, 23:09:02	0
simulation-task-005	Completed	26 Sept 2026, 23:09:02	0
simulation-task-006	Completed	26 Sept 2026, 23:09:02	0
simulation-task-007	Completed	26 Sept 2026, 23:09:02	0
simulation-task-008	Completed	26 Sept 2026, 23:09:02	0
simulation-task-009	Completed	26 Sept 2026, 23:09:02	0
simulation-task-010	Completed	26 Sept 2026, 23:09:02	0
simulation-task-011	Completed	26 Sept 2026, 23:09:02	0

Figure 13. Batch task execution state in the Azure portal.

7

Monitor Workload Completion

Use the Tasks view to monitor the number of submitted tasks and their states. The supplied execution evidence reports approximately 300 tasks and a 100% completion state, providing confirmation that the workload reached the completed stage.

Task	State	Created	Exit code
simulation-task-001	Completed	26 Sept 2026, 23:09:02	0
simulation-task-002	Completed	26 Sept 2026, 23:09:02	0
simulation-task-003	Completed	26 Sept 2026, 23:09:02	0
simulation-task-004	Completed	26 Sept 2026, 23:09:02	0
simulation-task-005	Completed	26 Sept 2026, 23:09:02	0
simulation-task-006	Completed	26 Sept 2026, 23:09:02	0
simulation-task-007	Completed	26 Sept 2026, 23:09:02	0
simulation-task-008	Completed	26 Sept 2026, 23:09:02	0
simulation-task-009	Completed	26 Sept 2026, 23:09:02	0
simulation-task-010	Completed	26 Sept 2026, 23:09:02	0
simulation-task-011	Completed	26 Sept 2026, 23:09:02	0

Figure 14. Azure Batch shows the workload at 100% completion.

8

Verify Final Task Status and Exit Codes

Check the final task records. In the supplied evidence, tasks 284 through 300 are displayed as Completed with exit code 0. This provides task-level confirmation for the displayed final tasks.

Home > engsimbatch2026neha | Jobs > simulation-job

simulation-job | Tasks engsimbatch2026neha

Search Add Add (JSON editor) Columns Refresh

Overview General Properties Environment settings Metadata **Tasks** Preparation tasks Release tasks Settings

simulation-task-284	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-285	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-286	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-287	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-288	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-289	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-290	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-291	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-292	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-293	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-294	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-295	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-296	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-297	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-298	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-299	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-300	Completed	26 Sept 2026, 23:11:50	0	...

Figure 15. Final task range 284–300 showing Completed and exit code 0.

9

Finish the Azure Execution Cycle

After task completion is verified, the Batch job contains the completed workload and the execution evidence can be recorded for the project. Outputs and logs can then be retained according to the project storage and reporting process.

Home > engsimbatch2026neha | Jobs > simulation-job

simulation-job | Tasks engsimbatch2026neha

Search Add Add (JSON editor) Columns Refresh

Overview General Properties Environment settings Metadata **Tasks** Preparation tasks Release tasks Settings

simulation-task-284	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-285	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-286	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-287	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-288	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-289	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-290	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-291	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-292	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-293	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-294	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-295	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-296	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-297	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-298	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-299	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-300	Completed	26 Sept 2026, 23:11:50	0	...

Figure 16. Final Azure task status used as execution evidence.

3. Combined Project Process Flow

The two implementations are connected at the workload level but are documented as separate process tracks: SimFlow demonstrates the application workflow, while Azure Batch demonstrates cloud task orchestration and execution.

SIMFLOW + AZURE BATCH — PROCESS FLOW

A. SimFlow prototype process



B. Azure Batch execution process



Figure 17. End-to-end process flow for the SimFlow prototype and Azure Batch execution.

Stage	SimFlow	Azure Batch	Output / Evidence
1	Create simulation	Prepare task workload	Simulation record
2	Create jobs	Create Batch job	Job/task grouping
3	Start workload	Queue tasks	Tasks ready for execution
4	Monitor	Assign tasks to nodes	Running/completed states
5	Complete	Verify task completion	Completed workload
6	Review results	Retain task evidence/logs	Results + execution proof

Process note: The prototype process and Azure Batch process are intentionally shown separately. The SimFlow interface demonstrates workload creation, status tracking and result review; the Azure section demonstrates Batch account, pool, job, task submission, monitoring and completion verification.